

ORIGINAL PAPER

Lookahead-Based Lookaside Selection: Mitigating Inline
Frequency Interference Between Satellite ConstellationsHuiliang Liu^{1,2} | Fei Peng^{1,2} | Zhe Zhang¹ | Yao Chu³¹Institute of Telecommunication and Navigation Satellites, China Academy of Space Technology, Beijing, China | ²Innovation Center of Satellite Communication System, China National Space Administration, Beijing, China | ³Department of Precision Instrument, Tsinghua University, Beijing, China

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ABSTRACT

The densification of non-geostationary orbit (NGSO) constellations has made co-frequency interference (CFI) a critical bottleneck for system capacity, particularly inline interference events where angular separation between heterogeneous satellite systems approaches zero. Existing mitigation strategies, such as spatial isolation or reactive handovers, often suffer from myopic decision-making, resulting in either persistent interference or excessive signaling overhead. This paper proposes a proactive lookahead-based lookaside selection algorithm to mitigate inline CFI in dynamic NGSO environments. By modeling satellite selection as a receding horizon optimization problem, the proposed algorithm integrates multi-step trajectory prediction, a temporal discount factor, and explicit switching penalties to optimize the long-term radio frequency environment. Simulation results using Walker-type constellations over a 30-day period demonstrate that the proposed method effectively eliminates severe interference events. At high latitudes (60°N), the probability of angular separation falling below 5° is reduced to 0%, compared to 1.1% for conventional highest-elevation strategies. Similar robustness is observed at lower latitudes and in challenging co-inclined “chasing” scenarios. Crucially, this performance is achieved with a balanced operational cost: Daily handovers increase by only 64% relative to stability-first baselines, while remaining 70% lower than elevation-maximization strategies. Parametric analysis further reveals that the optimal lookahead horizon correlates strongly with the average satellite pass duration, establishing a design guideline for proactive interference management in future 5G/6G-integrated non-terrestrial networks.

1 | Introduction

The rapid proliferation of non-geostationary orbit (NGSO) satellite constellations is fundamentally reshaping the landscape of global telecommunications [1]. Driven by the promise of ubiquitous coverage, low latency, and high-throughput broadband services, initiatives such as SpaceX's Starlink (planned up to 42,000 satellites), OneWeb (8368 satellites), Telesat (1671 satellites), and Amazon's Kuiper (3236 satellites) represent a paradigm shift, deploying vast numbers of satellites predominantly in low Earth orbit (LEO) [2]. These constellations aim to complement and integrate with terrestrial networks, forming key components of 5G and 6G infrastructures and enabling diverse applications

from global internet access to the Internet of Things (IoT) [3]. However, this unprecedented growth within finite orbital and spectral resources inevitably leads to significant challenges in spectrum coexistence, most notably the heightened risk of co-frequency interference (CFI) among these large-scale, heterogeneous NGSO systems operating in shared bands like Ku, Ka, and V [4].

Unlike interference scenarios involving geostationary orbit (GSO) systems, CFI between NGSO constellations presents unique and complex difficulties [5]. The high orbital velocities and diverse orbital parameters (altitudes, inclinations) inherent in NGSO systems result in highly dynamic network

topologies and constantly changing geometric relationships between satellites and ground stations (GSs) [6]. This dynamism leads to frequent, intense, and often asymmetric interference events that can severely degrade system performance, reduce communication capacity, and impact service availability [7]. A particularly critical manifestation of CFI is inline interference, occurring when a satellite from an interfering constellation passes close to the line-of-sight between a victim GS and its serving satellite. During such events, the small angular separation can lead to high antenna gain coupling, resulting in significantly elevated interference-to-noise ratios (I/N) and potential disruption of the communication link. The severity of inline events is influenced by factors including orbital diversity, user distribution patterns, and the complex aggregate interference from potentially thousands of interfering satellites. Quantifying the impact often involves metrics such as aggregate I/N, signal-to-interference-plus-noise ratio (SINR) degradation, percentage of throughput degradation (%DTp) under adaptive coding and modulation (ACM), system-level capacity loss, and, in the context of GSO protection, equivalent power flux density (EPFD) [8].

Significant research has focused on interference mitigation techniques, primarily developed for NGSO-GSO coexistence or terrestrial networks, but their direct application to inter-NGSO scenarios often reveals limitations [9]. Spatial isolation methods aim to increase the angular separation between desired and interfering signals [10]. Techniques include optimizing satellite antenna beam pointing, employing dynamic exclusion zones, adjusting satellite body tilting or pitch, defining minimum spatial isolation angles for GS access, or implementing arc avoidance strategies [11]. However, these spatial approaches can increase handover frequency, introduce latency, create coverage gaps, or require complex hardware like adaptive beamforming. Frequency domain isolation, such as band splitting, directly reduces available bandwidth and thus maximum achievable throughput. Power control mechanisms, including range-based, traffic-aware, or cognitive approaches, can help manage interference levels but are constrained by satellite onboard energy limitations and may be insufficient during severe inline events [12]. Broader radio resource management (RRM) strategies, encompassing joint allocation of power, spectrum, time, and beams, face complex, often non-convex optimization challenges exacerbated by the network's dynamism [13, 14]. Advanced approaches using deep learning or reinforcement learning for RRM or channel state information (CSI) prediction are also being explored but may involve significant computational complexity or rely on potentially delayed CSI. Accurate modeling of constellation distribution and interference probabilities is also crucial for effective analysis and mitigation design [15]. Existing techniques often lack proactive, long-term planning capabilities suitable for the fast-changing inter-NGSO environment, frequently resulting in reactive measures with substantial overhead or performance trade-offs. While spatial isolation techniques focus on physical separation, other approaches address interference through link adaptation. For instance, the latency limitations of reactive adaptive coding and modulation (ACM) loops in dynamic LEO environments have recently been highlighted [16]. A proactive ACM (P-ACM) scheme has been proposed, which leverages the deterministic nature of satellite trajectories to preemptively adjust modulation and coding schemes (MCS) before interference

occurs. This work exemplifies a critical paradigm shift in NGSO interference management: from reactive compensation to proactive prevention based on trajectory prediction. Building on this predictive philosophy, our study extends the “proactive” concept to the spatial domain, employing lookahead logic to optimize satellite handover sequences and ensure sustained angular separation over time.

While much focus is placed on interference between distinct NGSO systems, intra-system interference presents a parallel challenge. Many modern constellations, such as Starlink, are designed with multiple orbital shells featuring different altitudes and inclinations yet often share frequency bands [17]. Managing co-frequency operations across all shells simultaneously represents a highly dynamic and intricate resource allocation problem [18]. A pragmatic approach involves designating certain shells as primary operators while others actively avoid causing interference, simplifying the RRM complexity [8, 19]. Consequently, the victim and interferer framework investigated in this paper is applicable not only to inter-system coexistence but also to managing interference between different shells within the same multi-layer constellation. In this context, the “victim” system (or shell) employs avoidance strategies against interference from the “interferer” system (or shell).

In 5G communications, conditional handover (CHO) marks a paradigm shift by delegating handover decision-making from centralized network controllers to user equipment at the edge—enabling devices to proactively evaluate network conditions (e.g., signal strength, latency, QoS requirements) and autonomously determine optimal actions, unlike traditional fixed-policy-driven mechanisms [20]. Aligned with 6G's decentralization and intelligence principles as well as AI-driven Radio Access Networks (AI-RAN) goals, CHO integrates AI/ML and multimodal data fusion (from KPIs, traffic prediction models, external sources) to achieve near-real-time context-aware optimization, reducing congestion, minimizing handover failures, and enhancing seamless connectivity across heterogeneous networks [21, 22]. Building on this user-centric philosophy, the present work extends CHO with a novel user-side access selection strategy that leverages local context and intelligence at the GS to optimize resource utilization and service quality (while maintaining 5G/6G compatibility), whose effectiveness is demonstrated through extensive simulation and analytical evaluation [23].

This paper introduces a novel approach to mitigate inline CFI between NGSO constellations by enhancing the lookaside selection technique with multi-step lookahead planning. The fundamental lookaside principle involves a GS selecting an alternative visible satellite that maximizes the angular separation from potential interferers, leveraging the spatial diversity inherent in dense constellations. However, a purely reactive or myopic lookaside decision can lead to frequent, unnecessary handovers (switches) or suboptimal choices that offer only short-term relief. Our proposed method addresses this limitation by incorporating a lookahead mechanism that plans multiple time steps into the future. By evaluating potential satellite switch sequences over a defined horizon and employing a discount factor (γ) to balance immediate versus future separation angles, the algorithm seeks to optimize the

long-term interference environment. Furthermore, explicit switch penalties are integrated into the decision-making process to discourage excessive handovers, promoting link stability. The hemispherical sky view from the GS is discretized using a hexagonal grid structure for efficient spatial analysis. We simulate representative Walker-type constellations to evaluate the effectiveness of our lookahead-based lookaside algorithm. Performance is assessed using metrics relevant to inline interference avoidance, specifically the cumulative probability of angular separation falling below critical thresholds (e.g., 5° and 10°) and the total number of satellite switches over the simulation period.

The key contribution of this work is the proposal and formulation of a lookahead-based lookaside satellite selection algorithm, which incorporates multi-step planning, a discount factor, and switch penalties for proactive inline interference mitigation in dynamic inter-NGSO or intra-NGSO scenarios. To validate this approach, we develop a simulation framework using realistic constellation models and sky discretization to evaluate the long-term performance of the proposed technique. Through this framework, we conduct a comparative analysis demonstrating the superiority of the lookahead approach in maximizing angular separation and balancing handover frequency compared to baseline strategies, such as simple highest-elevation or longest-access-time selection.

The remainder of this paper is organized as follows: Section 2 details the system model, including constellation parameters, channel characteristics, and the inline interference scenario. Section 3 presents the proposed lookahead-based lookaside algorithm, including the planning horizon, discount factor, and switch penalty considerations. Section 4 provides comprehensive simulation results and performance analysis comparing our approach with baseline methods. Finally, Section 5 concludes the paper, summarizes key findings, and discusses potential directions for future research.

2 | System Model and Interference Scenarios

2.1 | Scenario Description

This section details the system model for co-frequency interference between NGSO constellations and the specific interference scenarios considered. We focus on the inline (co-located) scenario, where the ground user terminal interacts with both the victim and interferer systems, representing the most severe interference case due to minimal angular separation and heightened signal degradation [1].

The scenario involves two heterogeneous NGSO constellations operating in the same frequency bands (e.g., Ka/Ku), with a co-located ground station (GS) serving as the common user terminal. The victim constellation is the protected system, while the interferer is the aggressor. Inline interference occurs when the GS, victim satellite, and interferer satellite are nearly aligned in line-of-sight, resulting in small isolation angles and high I/N ratios that can cause outages [1]. Figure 1 provides a schematic of this scenario, illustrating the GS at latitude 30°N, longitude 0°, and the satellites in their respective orbits. The isolation angle α

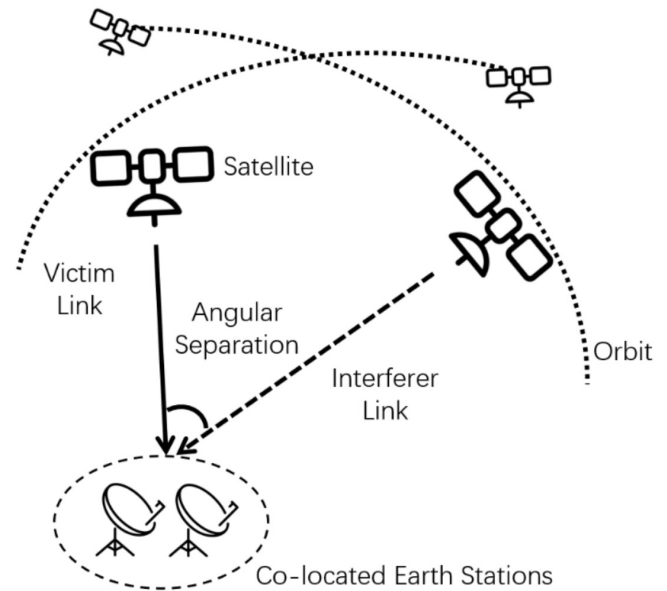


FIGURE 1 | Schematic of the interference scenario.

between the selected victim and interferer satellites is defined as the angular separation from the GS perspective, computed using the spherical law of cosines:

$$\alpha = \arccos(\sin el_v \sin el_i + \cos el_v \cos el_i \cos \Delta az) \cdot (180^\circ / \pi)$$

$$\Delta az = |az_v - az_i|$$

where (az_v, el_v) and (az_i, el_i) are the azimuth and elevation of the victim and interferer satellites, respectively. Small α (e.g., $<5^\circ$) indicate severe interference, as beams overlap significantly [1]. Figure 2 illustrates a schematic diagram of the distribution of an inclined orbit constellation and a near-polar orbit constellation within the visible range of a GS at 60°N at a specific moment. In the diagram, green circles represent the victim system, orange squares represent the interferer system, and yellow indicates the selected working satellite at that moment.

The critical isolation angle threshold is determined based on the reference radiation patterns for earth station antennas outlined in Appendix 8 of the Radio Regulations. To simulate a worst-case interference scenario involving small-aperture user terminals (e.g., diameter-to-wavelength ratios of 13.0, corresponding to a 39 cm antenna in Ku-band), we assume an antenna with a maximum gain of 30 dBi and a beamwidth of 5.3°. Under this configuration, the antenna gain decreases by 10.7 dB relative to the peak at an off-axis angle of 5°. Therefore, inline events with angular separations below 5° generally result in a carrier-to-interference ratio (C/I) below 10.7 dB (assuming comparable power flux-density), which exceeds typical system design margins and is thus classified as severe interference.

2.2 | System Parameters

The constellations are modeled using Walker construction, with parameters summarized in Table 1. Simulations run from September 17, 2025, 00:00 UTC to October 16, 2025, 23:59 UTC

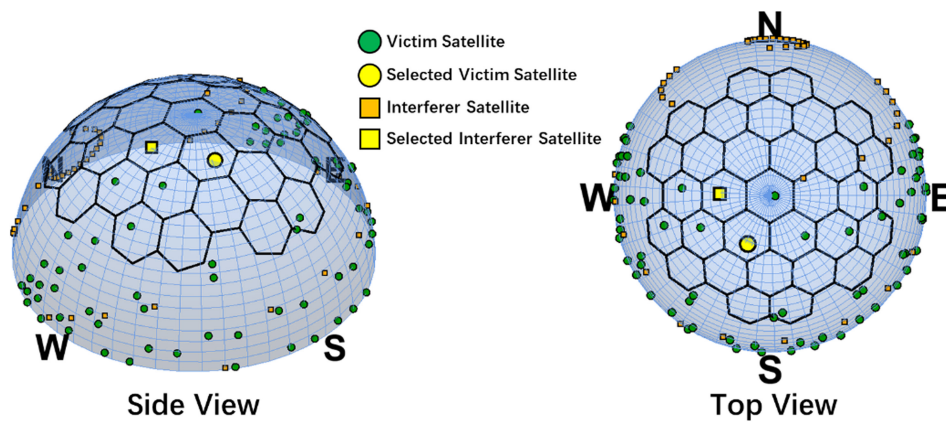


FIGURE 2 | Schematic of the constellations in the visible range of ground station.

TABLE 1 | Constellation parameters.

Parameter	Victim system	Interferer system
Number of planes	32	9
Satellites per plane	25	20
Altitude (km)	1150	1200
Inclination (°)	60	88
RAAN0 (°)	0	0
Constellation type	Walker Delta	Walker Star
Minimum elevation (°)	40	40
Selection strategy	Highest_elevation, longest_access_time, lookahead	Longest_access_time

(30 days), discretized at 1-min intervals (43,200 steps). The ground station is deliberately located at 0° E, 60° N with an altitude of 0 km. In this study, the ground station is strategically positioned at 60° N to situate it at the edge of the victim constellation's service area, where nearly half of the sky dome above the zenith lacks visible victim satellites. Under such constrained conditions, opportunities to “bypass” the interferer are more limited, thereby increasing the difficulty of optimizing satellite selection strategies to achieve the objective of evading inline interference. To further validate the universality of the proposed approach, performance evaluations at 40° N and 20° N ground station locations are also presented and analyzed throughout this paper, demonstrating the method's robust applicability across diverse geographic deployments ranging from high-latitude to near-equatorial regions. The hemispherical field of view from the GS is discretized into a hexagonal grid for efficient computation of obstructed regions and aggregation of visibility statistics. Each hexagonal cell has a side length of $r_{cell} = 10^\circ$, chosen to balance resolution and computational efficiency while covering the working area (elevations $\geq 40^\circ$). This discretization approximates the continuous sky dome into 31 cells within the maximum co-elevation radius ($\rho_{max} = 50^\circ$).

It is important to note that while the primary results do not explicitly detail performance at the Equator (0° N), the proposed lookahead strategy is equally applicable and effective in such regions. At 0° N, the dominant operational constraint for NGSO systems transitions to safeguarding the geostationary (GSO) arc, typically enforced via an exclusion zone—a simplified

East–West band across the zenith prohibiting transmissions to prevent GSO interference. This zone creates a constrained field of view (FoV) that mirrors the geometric restrictions at 60° N, where satellite visibility is curtailed north of the orbital inclination limit. In equatorial deployments, the lookahead algorithm would function within the allowable scanning areas outside the GSO exclusion zone, applying identical multi-step prediction to optimize angular separations—analogueous to its handling of the southern-sky constraints at higher latitudes.

2.3 | Sky Discretization and Coverage Representation

To systematically analyze satellite visibility and selection patterns, we discretize the hemispherical sky into hexagonal cells. This tessellation provides uniform spatial coverage while minimizing geometric distortion compared to rectangular grids.

The sky hemisphere is represented in a planar polar coordinate system (ρ, θ), where:

$$\begin{aligned}\rho &= 90^\circ - el \\ \theta &= 90^\circ - az\end{aligned}$$

Here, ρ represents the zenith angle (distance from zenith), and θ is measured counter-clockwise from the East direction. Converting to Cartesian coordinates (X, Y):

$$X = \rho \cos(\theta)$$

$$Y = \rho \sin(\theta)$$

Each hexagonal cell has a side length of $r_{\text{cell}} = 10^\circ$, chosen to balance resolution and computational efficiency while covering the working area. The centers of adjacent cells are separated by $d = r_{\text{cell}}\sqrt{3} \approx 17.32^\circ$. The hexagonal lattice basis vectors are:

$$\mathbf{v}_1 = [d, 0]$$

$$\mathbf{v}_2 = [d/2, d\sqrt{3}/2]$$

Cell centers (X_c, Y_c) are located at integer combinations: $(X_c, Y_c) = m\mathbf{v}_1 + n\mathbf{v}_2$ for integers m, n , subject to $\rho_c = \sqrt{X_c^2 + Y_c^2} \leq 50^\circ$ (corresponding to the minimum elevation constraint of 40°). This discretization approximates the continuous sky dome into 31 cells within the maximum co-elevation radius ρ_c .

It is worth noting that this hexagonal discretization serves purposes beyond immediate geometric calculation. Primarily, it facilitates the visualization of long-term statistical distributions of satellite selections. More importantly, it structures the continuous sky dome into a finite state space, which is a prerequisite for extending this framework to reinforcement learning algorithms in future work, thereby preventing the dimensionality explosion associated with continuous state spaces. Furthermore, this cellular approach aligns with emerging concepts in 6G non-terrestrial networks, potentially offering a spatial coding framework analogous to terrestrial tracking area codes for fine-grained resource management.

2.4 | Baseline Satellite Selection Strategies

The interferer uses the `longest_access_time` strategy to simulate a stability-focused aggressor. For the victim, we evaluate baseline strategies (`highest_elevation` and `longest_access_time`) before introducing the lookahead strategy in Section 3.

The **highest_elevation** strategy selects the satellite with the maximum elevation angle at each observation time:

$$(\text{plane}^*, \text{sat}^*) = \arg \max_{(p,s) \in \mathcal{V}(t)} el_{p,s}(t)$$

where $\mathcal{V}(t)$ is the set of visible satellites above the minimum elevation threshold at time t .

The **longest_access_time** strategy prioritizes satellite persistence to minimize handover overhead. At each decision epoch, if the currently selected satellite remains visible above the minimum elevation, it is retained. Otherwise, the algorithm selects the satellite with the longest predicted future visibility:

$$(\text{plane}^*, \text{sat}^*) = \arg \max_{(p,s) \in \mathcal{V}(t)} \Delta T_{p,s}(t)$$

where $\Delta T_{p,s}(t)$ is the remaining time until satellite (p,s) drops below the minimum elevation threshold, computed by forward propagation:

$$\Delta T_{p,s}(t) = \max \{ \tau : el_{p,s}(t + \tau) \geq el_{\min} \}$$

The `longest_access_time` strategy prioritizes satellite persistence to minimize handover overhead. At each decision epoch, if the currently selected satellite remains visible above the minimum elevation (40°), it is retained. Handovers are triggered exclusively when the serving satellite drops below the 40° elevation threshold or becomes obstructed; switches are not performed in response to interference levels or other dynamic factors.

2.5 | Simulation Results for Baseline Strategies

Simulation results for these baseline strategies are presented below, focusing on key metrics: expected satellite count (average visible satellites per hex cell), selection probability (fraction of time a cell is selected), cumulative distribution function (CDF) of angles, and cumulative switches.

The expected satellite count for each hex cell is defined as the average number of visible satellites (elevation $\geq 40^\circ$) projected into that cell over all simulation time steps. The selection probability for each hex cell is the fraction of time steps during which the selected satellite falls within that cell.

For `highest_elevation` strategy implemented by victim system, Figure 3 shows the expected satellite count distribution, concentrated in the southern cells of the GS field of view, due to the relation between inclination of the constellation and the GS latitude. Figure 4 depicts the selection probability distribution under the `highest_elevation` strategy for the GS at 60°N . The probabilities are strongly concentrated around the zenith, indicating a preference for satellites at the highest elevations, while peripheral hexagonal cells exhibit minimal selection rates due to the strategy's focus on instantaneous elevation maximization.

To demonstrate performance characteristics across three geographically diverse GS locations spanning 60°N , 40°N , and

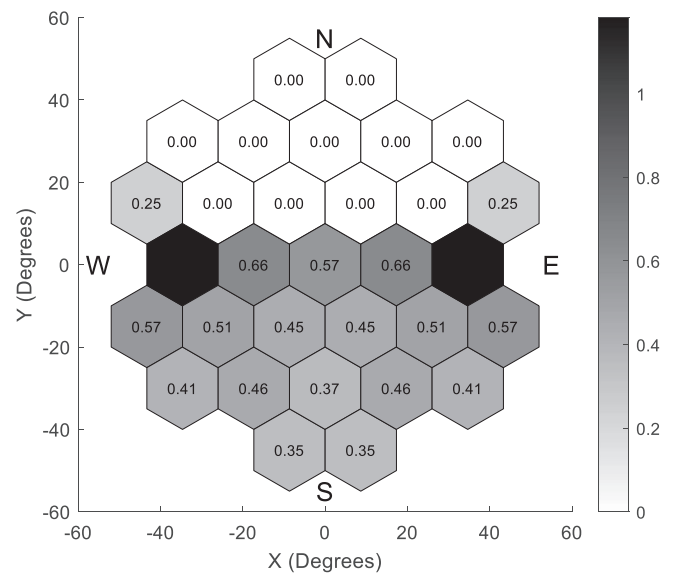


FIGURE 3 | Expected satellite count distribution of victim system with GS at 60°N .

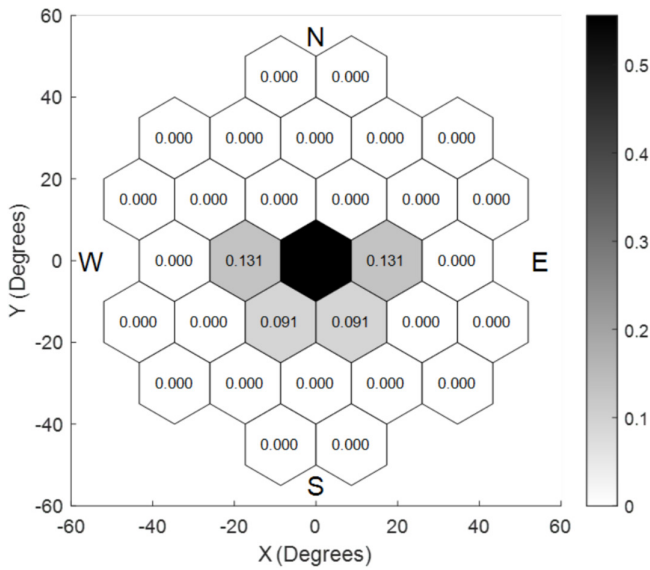


FIGURE 4 | Selection probability of highest_elevation strategy with GS at 60°N.

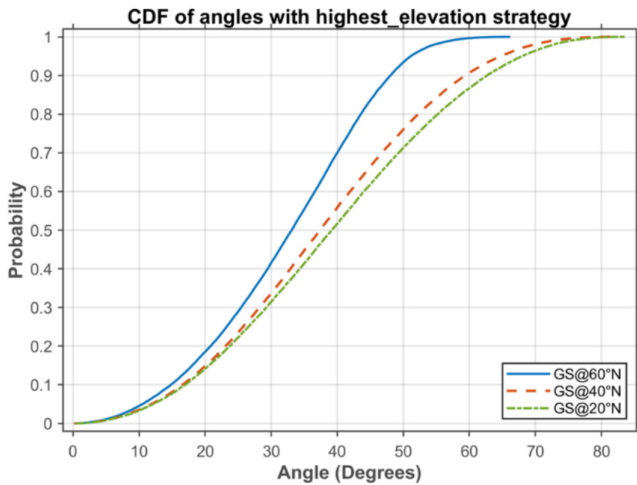


FIGURE 5 | CDF of angles with highest_elevation strategy.

20°N latitudes, Figure 5 presents the CDF of isolation angles and Figure 6 illustrates the cumulative satellite switches. Table 2 summarizes the comprehensive performance metrics over a 30-day simulation period.

The minimum isolation angles achieved are critically low across all latitudes, ranging from 0.21° (GS@60°N) to 0.40° (GS@40°N), indicating frequent occurrences of near-coincident victim and interferer satellite geometries. The maximum isolation angles exhibit strong latitude dependence, increasing from 66.20° at high latitude (60°N) to 83.51° at equatorial regions (20°N). This trend reflects the geometric diversity of satellite visibility patterns—higher latitudes experience more concentrated zenith crossings from inclined orbits, leading to frequent co-location of both constellations at high elevation angles.

The probability of severe interference ($P(\alpha < 5^\circ)$) ranges from 0.8% to 1.1%, with the highest occurrence at 60°N latitude. While these values represent relatively infrequent events in absolute

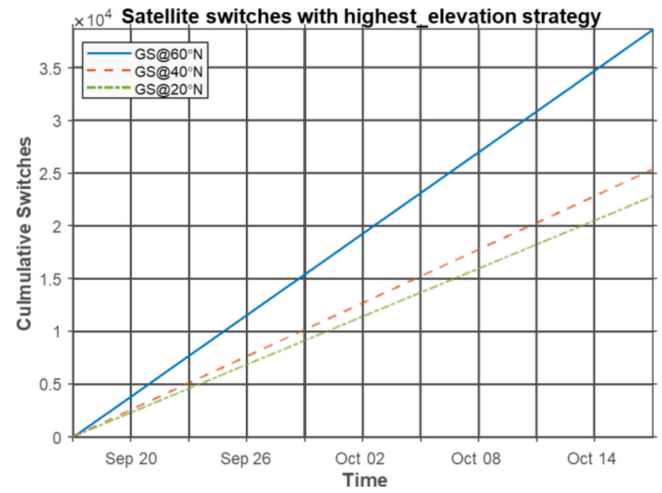


FIGURE 6 | Cumulative switches with highest_elevation strategy.

TABLE 2 | Performance of highest_elevation strategy.

	Min α	Max α	$P(\alpha < 5^\circ)$	$P(\alpha < 10^\circ)$	Switches/day
GS@60°N	0.21°	66.20°	1.1%	4.5%	1289
GS@40°N	0.40°	81.52°	0.9%	3.5%	846
GS@20°N	0.36°	83.51°	0.8%	3.3%	759

terms, they translate to approximately 11.5–15.8 min per day of critical interference exposure. Moderate interference ($P(\alpha < 10^\circ)$) occurs 3.3%–4.5% of the time, corresponding to 47–65 min daily. The highest interference probability occurs at 60°N (4.5%), suggesting that the myopic elevation-maximization criterion is particularly vulnerable in high-latitude deployments where satellite ground tracks converge.

The highest_elevation strategy exhibits extremely high handover rates, ranging from 759 switches/day (20°N) to 1289 switches/day (60°N). This behavior stems from the continuous tracking of the instantaneous highest-elevation satellite, that any time a rising satellite surpasses the current selection's elevation.

Figure 7 presents the selection probability distribution under the longest_access_time strategy for the GS at 60°N. Unlike the zenith-focused pattern of the highest_elevation approach, this strategy yields a more dispersed distribution across the visible sky, with higher probabilities in cells corresponding to longer satellite visibility windows, promoting extended connection durations over peak elevation. Figure 8 presents the CDF of isolation angles and Figure 9 illustrates the cumulative satellite switches. Table 3 summarizes the comprehensive performance metrics over a 30-day simulation period, emphasizing connection persistence over instantaneous link quality optimization.

Minimum isolation angles remain critically small (0.19°–0.38°), comparable to or slightly worse than the highest_elevation baseline. However, maximum isolation angles demonstrate substantial improvement, reaching 95.70°–99.48° across all latitudes.

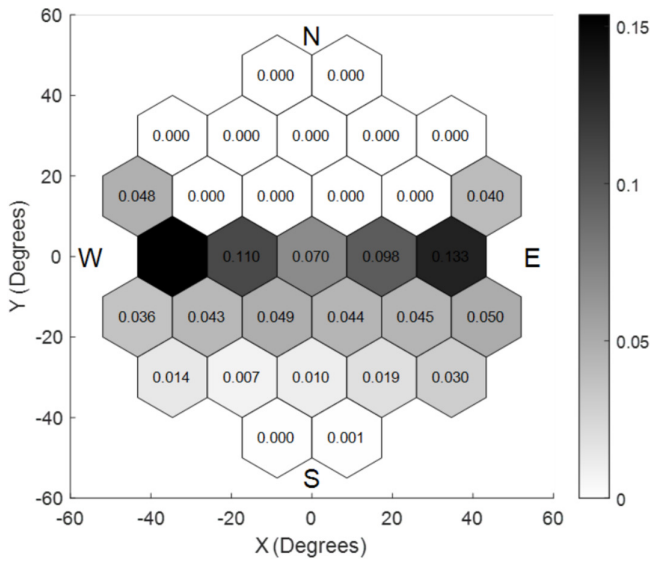


FIGURE 7 | Selection probability with longest_access_time strategy with GS at 60°N.

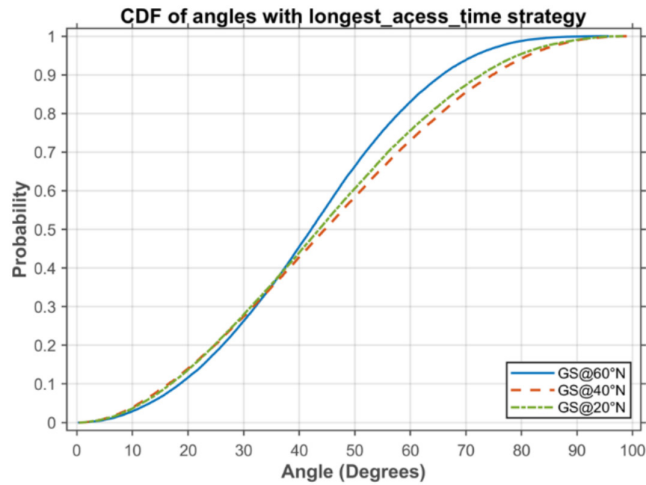


FIGURE 8 | CDF of angles with longest_access_time strategy.

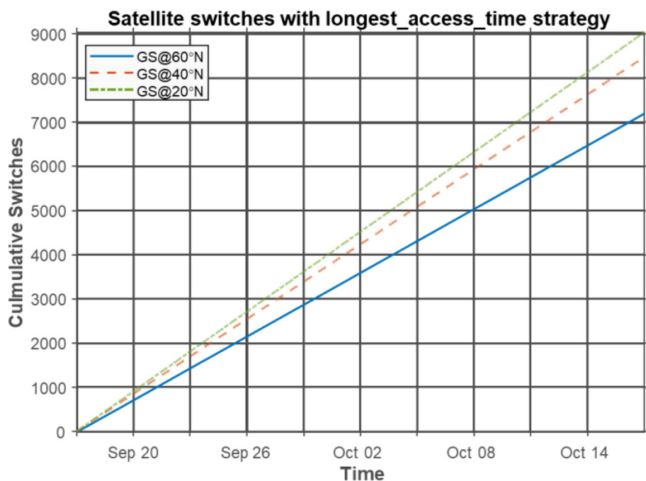


FIGURE 9 | Cumulative switches with longest_access_time strategy.

TABLE 3 | Performance of longest_access_time strategy.

	Min α	Max α	$P(\alpha < 5^\circ)$	$P(\alpha < 10^\circ)$	Switches/day
GS@60°N	0.19°	95.70°	0.7%	2.9%	240
GS@40°N	0.35°	99.48°	1.0%	3.9%	283
GS@20°N	0.38°	98.28°	0.9%	3.7%	301

This 14°–16° increase in maximum angular separation suggests that the strategy occasionally maintains satellites through geometrically favorable phases of their orbital passes, even as they descend from zenith.

The longest_access_time strategy does not consistently reduce interference probability compared to highest_elevation. At 60°N, $P(\alpha < 5^\circ)$ improves to 0.7%, but at 40°N and 20°N, interference probability increases to 1.0% and 0.9%, respectively. This counterintuitive behavior reveals a fundamental limitation: maintaining satellites through their entire visibility pass “locks in” the interference geometry for extended periods. If a satellite is selected during a phase of poor interferer separation, the system remains trapped in this unfavorable state for the duration of the pass. In contrast, highest_elevation’s frequent reselection provides opportunities to escape poor geometries, albeit at the cost of higher handover rates. Moderate interference ($P(\alpha < 10^\circ)$) shows similar mixed results, improving at 60°N but worsening at lower latitudes.

The strategy achieves its primary design objective: dramatic reduction in handover frequency to 240–301 switches/day—an 81% reduction compared to highest_elevation at 60°N (240 vs. 1289 switches/day). Mean satellite connection durations extend to approximately 4.8–6.0 min, closely matching typical LEO satellite pass durations for mid-to-high elevation angles.

The longest_access_time strategy successfully minimizes signaling overhead and service disruptions, making it suitable for handover-constrained systems with limited control channel capacity or latency-sensitive applications where connection continuity is paramount. However, its failure to efficiently reduce interference probability limits its applicability for robust coexistence in dense NGSO environments. This underscores the need for advanced strategies that minimize small-angle probabilities while balancing handover frequency, motivating the lookahead-based lookaside approach in Section 3.

3 | Lookahead-Based Satellite Selection Algorithm

3.1 | Algorithm Design Philosophy

To address the limitations of baseline strategies identified in Section 1, this section introduces the proposed lookahead-based lookaside selection algorithm. The lookaside technique enables the ground station (GS) to dynamically select an alternative visible satellite from the victim constellation that maximizes angular separation from the interferer’s selected satellite, thereby mitigating inline interference without requiring coordination between constellations.

Classical satellite selection strategies operate myopically, optimizing instantaneous metrics without considering the temporal evolution of interference geometry. The proposed **lookahead algorithm** addresses this limitation through three key design principles:

1. **Multi-step temporal reasoning:** Evaluate candidate satellites not only by current performance but also by their predicted interference isolation over a future time horizon, accounting for the orbital dynamics of both victim and interferer constellations.
2. **Proactive interference avoidance:** Integrate real-time interferer position awareness into the selection logic, preemptively avoiding regions where interference is likely to occur.
3. **Handover cost-benefit analysis:** Explicitly model the operational cost of satellite switching, creating a quantitative tradeoff between maintaining stable connections and pursuing better interference geometries

The algorithm adopts a receding horizon optimization framework, common in model predictive control, where decisions are made by solving a finite-horizon optimization problem at each time step, then executing only the immediate action before re-planning at the next epoch. The lookahead mechanism assumes knowledge of the interferer's handover policy for optimal prediction, but it is robust to partial or inferred information in non-cooperative scenarios. For cooperative multi-shell systems, this data is readily available. In frequency coordination among NGSO operators, exchanged parameters (e.g., selection strategies) serve as priors. For non-cooperative cases, victim systems with sensing capabilities can detect interferer beam patterns in hotspots, while others can infer strategies from accumulated interference data and public orbital elements, enabling probabilistic lookahead.

3.2 | Reward Function Design

Each satellite in the set is uniquely identified by the pair (p, s) , where p denotes the orbital plane index within the constellation, and s represents the specific satellite index within that orbital plane. At decision time t_k , let $S_k = \{(p_i, s_i)\}_{i=1}^{N_k}$ denote the set of N_k visible victim satellites above the minimum elevation threshold el_{min} . Let (p_{curr}, s_{curr}) represent the currently selected satellite. The interferer system independently selects satellite (p_I, s_I) using its own strategy (typically `longest_access_time`).

The **lookahead selection problem** is formulated as:

$$(\hat{p}, \hat{s}) = \arg \max_{(p,s) \in S_k^{allowed}} J(p, s, t_k)$$

where $S_k^{allowed} \subseteq S_k$ is the subset of satellites not located in obstructed sky regions, and $J(p, s, t_k)$ is the **total expected reward** over the lookahead horizon.

The total reward $J(p, s, t_k)$ for selecting satellite (p, s) at time t_k is computed as:

$$J(p, s, t_k) = R_0(p, s, t_k) - C_{switch}(p, s) + \sum_{h=1}^{H-1} \gamma^h \cdot R_h(p, s, t_k)$$

where:

- $R_0(p, s, t_k)$: Immediate reward at current time
- $C_{switch}(p, s)$: Switching cost penalty
- $R_h(p, s, t_k)$: Future reward at lookahead step h
- H : Lookahead horizon (number of time steps)
- $\gamma \in [0, 1]$: Temporal discount factor

1. Immediate Reward

The immediate reward quantifies the interference isolation benefit of selecting satellite (p, s) at time t_k :

$$R_0(p, s, t_k) = \ln(\alpha(p, s, p_I, s_I, t_k) + \epsilon)$$

where $\alpha(\cdot)$ is the isolation angle (in degrees) between victim satellite (p, s) and interferer satellite (p_I, s_I) . And $\epsilon = 1e^{-5}$. Larger isolation angles yield higher rewards, incentivizing selections that maximize angular separation from the interferer. The use of the natural logarithm ($\ln$) is intentional. It applies the principle of diminishing marginal utility to the angular separation. This function ensures that the reward gain is substantial when increasing a small, critical separation angle, which is crucial for avoiding severe interference. In contrast, the reward gain for increasing an already large, safe angle is significantly smaller. This allows the algorithm to prioritize the avoidance of highly disruptive, small-angle events over minor optimizations in an already stable geometry. The reward function employs a conservative worst-case model, assuming active co-frequency users in interferer beams for hotspot scenarios, to prioritize robust avoidance. This may trigger precautionary handovers, but the switching penalty (β) prevents excess by requiring substantial predicted gains. For sparse regions with spectrum availability, complementary frequency-domain strategies could be integrated, though this is beyond the current geometric focus.

For multiple simultaneous interferers (e.g., overlapping beams or sidelobes), the reward is adapted to $R_0(p, s, t_k) = \ln\left(\min_{m \in \{1, 2, \dots, M\}} \alpha(p, s, p_{I,m}, s_{I,m}, t_k) + \epsilon\right)$ across all M active interferers, prioritizing avoidance of the dominant (smallest-angle) source. $\alpha(p, s, p_{I,m}, s_{I,m}, t_k)$ is the isolation angle between the candidate victim satellite (p, s) and the m -th interferer. This conservative aggregation maintains the algorithm's focus on severe inline risks without significant complexity increase.

2. Switching Cost

To discourage excessive handovers, a fixed penalty is imposed when selecting a different satellite from the current one:

$$C_{switch}(p, s) = \begin{cases} \beta & \text{if } (p, s) \neq (p_{curr}, s_{curr}) \\ 0 & \text{otherwise} \end{cases}$$

where β is the **switching penalty parameter**. Typical values range from 0 to 5, representing the equivalent isolation angle loss acceptable in exchange for maintaining connection stability.

This formulation naturally balances exploration (seeking better satellites) and exploitation (maintaining good connections). A higher β promotes satellite persistence, while lower values enable more aggressive interference avoidance.

3. Future Rewards

For each lookahead step $h \in \{1, 2, \dots, H-1\}$, the future reward R_h evaluates the predicted isolation angle at time $t_k + h\Delta t$:

$$R_h(p, s, t_k) = \begin{cases} \ln(\alpha(p, s, p_I, s_I, t_k + h\Delta t) + \epsilon) & \text{if feasible} \\ 0 & \text{otherwise} \end{cases}$$

where:

- Δt is the lookahead time step (e.g., 1 min)
- $(p_I^{(h)}, s_I^{(h)})$ is the predicted interferer satellite selection at $t_k + h\Delta t$
- Feasibility requires (p, s) remains above el_{min} at $t_k + h\Delta t$

The **temporal discount factor** γ encodes uncertainty in future predictions. Values near 0.9 balance near-term and medium-term considerations; $\gamma = 1$ treats all future rewards equally (undiscounted), while $\gamma \rightarrow 0$ recovers the myopic immediate-reward strategy.

If at any lookahead step h the candidate satellite becomes infeasible (drops below minimum elevation), all subsequent future rewards are set to zero, discouraging selection of satellites near the end of their visibility window.

Algorithm 1 presents the complete lookahead satellite selection procedure executed at each decision epoch t_k .

3.3 | Computational Complexity

The algorithm's computational complexity is analyzed per decision epoch:

1. **Candidate enumeration:** Iterating over N_k visible satellites: $O(N_k)$
2. **Lookahead evaluation:** For each candidate, the lookahead loop executes up to $H-1$ steps, each requiring:
 - Orbital propagation: $O(1)$ for simplified Keplerian dynamics
 - Interferer prediction: $O(N_I)$ where N_I is the number of visible interferer satellites
 - Isolation angle computation: $O(1)$

Total per candidate: $O(H \cdot N_I)$

3. **Overall complexity:** $O(N_k \cdot H \cdot N_I)$

For typical LEO constellations, $N_k \approx 5-15$ visible satellites, horizon $H = 3-5$ steps, and $N_I \approx 5-15$, yielding $O(10^2)-O(10^3)$ operations per decision epoch. With decision intervals of 1-10 min

ALGORITHM 1: Lookahead-Based Satellite Selection

Input:

- Current time t_k
- Visible satellite set $\mathcal{S}_k = \{(p_i, s_i)\}_{i=1}^{N_k}$
- Current selection (p_{curr}, s_{curr})
- Lookahead horizon H , time step Δt
- Discount factor γ , switching penalty β
- Constellation parameters (both victim and interferer)

Output:

- Selected satellite $(\hat{p}, \hat{s})$

```

1. Initialize:  $J_{max} \leftarrow -\infty$ ,  $(\hat{p}, \hat{s}) \leftarrow (-1, -1)$ 
2. Predict current interferer selection:
    $(p_I^{(0)}, s_I^{(0)}) \leftarrow f_{interferer}(t_k)$ 
3. Filter candidate satellites:
    $\mathcal{S}_k^{allowed} \leftarrow \mathcal{S}_k$ 
4. For each  $(p, s) \in \mathcal{S}_k^{allowed}$ :
   a. Compute immediate reward:
       $R_0 \leftarrow \ln(\alpha(p, s, p_I^{(0)}, s_I^{(0)}, t_k) + \epsilon)$ 
   b. Compute switching cost:
      if  $(p, s) \neq (p_{curr}, s_{curr})$  then
          $C_{switch} \leftarrow \beta$ 
      else
          $C_{switch} \leftarrow 0$ 
   c. Initialize total reward:
       $J \leftarrow R_0 - C_{switch}$ 
   d. Lookahead loop (for  $h = 1$  to  $H-1$ ):
      i. Propagate satellite position:
          $(az^{(h)}, el^{(h)}) \leftarrow \text{position of } (p, s) \text{ at } t_k + h\Delta t$ 
      ii. Check visibility constraint:
         if  $el^{(h)} < el_{min}$  then
            break (exit lookahead loop)
      iii. Check obstruction constraint:
      iv. Predict interferer selection:
          $(p_I^{(h)}, s_I^{(h)}) \leftarrow f_{interferer}(t_k + h\Delta t)$ 
      v. Compute future reward:
          $R_h \leftarrow \ln(\alpha(p, s, p_I^{(h)}, s_I^{(h)}, t_k + h\Delta t) + \epsilon)$ 
      vi. Accumulate discounted reward:
          $J \leftarrow J + \gamma^h \cdot R_h$ 
   e. Update best selection:
      if  $J > J_{max}$  then
          $J_{max} \leftarrow J$ 
          $(\hat{p}, \hat{s}) \leftarrow (p, s)$ 
5. Return  $(\hat{p}, \hat{s})$ 

```

and modern processors executing 10^9 operations per second, real-time implementation is easily achievable.

4 | Simulation Results and Performance Analysis

This section presents the simulation results for the proposed lookahead-based lookaside selection algorithm, evaluating its effectiveness in mitigating inline frequency interference. Simulations were conducted over a 30-day period from September 17, 2025, 00:00 UTC to October 16, 2025, 23:59 UTC, discretized at 1-min intervals (1440 steps per day, totaling 43,200 steps). The ground station is located at 0°E , 60°N , 0°E , 40°N and 0°E , 20°N with an altitude of 0 km.

4.1 | Simulation Results

The proposed lookahead strategy demonstrates transformative performance improvements across all metrics and geographic locations, as presented in Table 4. The algorithm employs adaptive parameter configurations (H , β , γ) selected for each latitude, reflecting its flexibility in balancing competing operational objectives.

The selected lookahead horizon H decreases from 5 steps (60°N) to 4 steps (40°N) to 3 steps (20°N), correlating with satellite visible duration. High-latitude stations benefit from longer prediction windows to anticipate interference evolution across the whole visible duration of a single satellite, while equatorial stations with relatively lower satellite diversity achieve sufficient performance with shorter horizons, also reducing computational burden. Conversely, the switching penalty β increases from 2° (60°N) to 4° (20°N), directly correlating with baseline handover rates. Equatorial deployments require stronger persistence incentives to prevent excessive reselection. The uniform discount factor $\gamma=0.5$ across all latitudes indicates that emphasizing immediate over distant-future rewards provides robust performance regardless of geographic location.

To illustrate performance characteristics across three geographically diverse GS locations spanning 60°N, 40°N, and 20°N latitudes, Figure 10 presents the CDF of isolation angles and Figure 11 illustrates the cumulative satellite switches.

The lookahead strategy achieves a dramatic increase in minimum isolation angles: 6.07° (60°N), 3.31° (40°N), and 1.91° (20°N). The worst-case interference geometries are effectively eliminated through the algorithm's proactive avoidance of satellite selections that will lead to small future isolation angles. Maximum isolation angles remain comparable to or exceed the longest_access_time strategy, demonstrating that forward prediction does not sacrifice geometric favorability. The algorithm successfully identifies and maintains connections with satellites offering sustained large angular separations.

The most significant performance gain manifests in critical interference probability reduction across all three geographic locations. At 60°N latitude, the lookahead strategy achieves complete elimination of severe interference events with $P(\alpha < 5^\circ) = 0\%$, compared to 1.1% for highest_elevation and 0.7% for longest_access_time, representing a 100% reduction. Moderate interference is similarly suppressed to $P(\alpha < 10^\circ) = 0.16\%$, versus 4.5% and 2.9% for the baseline strategies, corresponding to 96% and 94% reductions respectively. At 40°N latitude, critical interference probability drops to 0.0048%, compared to 0.9% for both

baseline approaches, achieving a 99.5% reduction, while moderate interference decreases to 0.15% from 3.5% and 3.9%, representing 96% reductions. Even at the challenging 20°N equatorial location, the lookahead algorithm maintains $P(\alpha < 5^\circ) = 0.039\%$, compared to 0.8% and 0.9% for the baselines (95% reduction), and reduces moderate interference to 0.35% from 3.3% and 3.7% (89%–91% reduction). These results demonstrate the lookahead algorithm's consistent capability to maintain interference probability well below typical system requirements across diverse geographic deployments.

The lookahead strategy achieves a balanced handover profile of 344–393 switches per day, representing a 55–70% reduction compared to highest_elevation (759–1289 switches/day) and a 30–39% increase over longest_access_time (240–301 switches/day). Unlike baseline strategies, the lookahead approach exhibits modest latitude dependence with only 13% variation, substantially lower than highest_elevation's 70% variation, indicating that adaptive parameter tuning successfully normalizes operational behavior across diverse geographic deployments and simplifies system-wide operational planning and resource allocation.

Figure 12 shows the selection probability distribution under the proposed lookahead strategy for the GS at 60°N. It demonstrates the different pattern compared with Figure 4 and Figure 7 of baseline strategies. The pattern reveals a strategic spread across multiple hexagonal cells, balancing interference avoidance with handover minimization, as the algorithm proactively selects regions offering sustained angular separation from interferers while incorporating future predictions and switching costs.

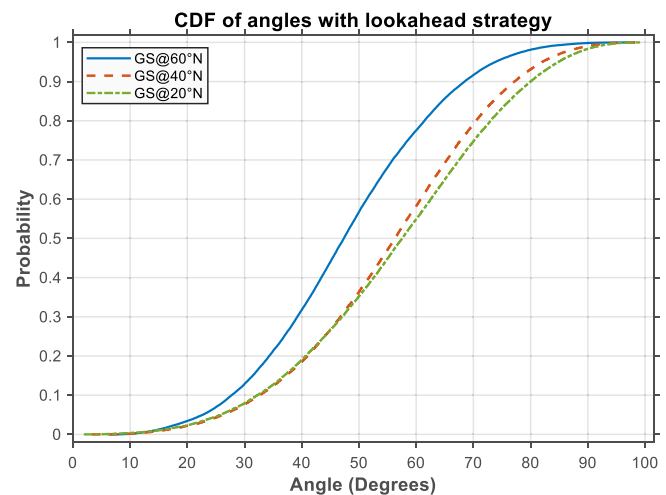


FIGURE 10 | CDF of angles with lookahead strategy.

TABLE 4 | Performance of lookahead strategy.

Parameters		Min α	Max α	$P(\alpha < 5^\circ)$	$P(\alpha < 10^\circ)$	Switches/day
GS@60°	$H=5 \beta=2 \gamma=0.5$	6.07°	98.25°	0%	0.16%	393
GS@40°	$H=4 \beta=3 \gamma=0.5$	3.31°	99.59°	0.0048%	0.15%	378
GS@20°	$H=3 \beta=4 \gamma=0.5$	1.91°	99.27°	0.039%	0.35%	344

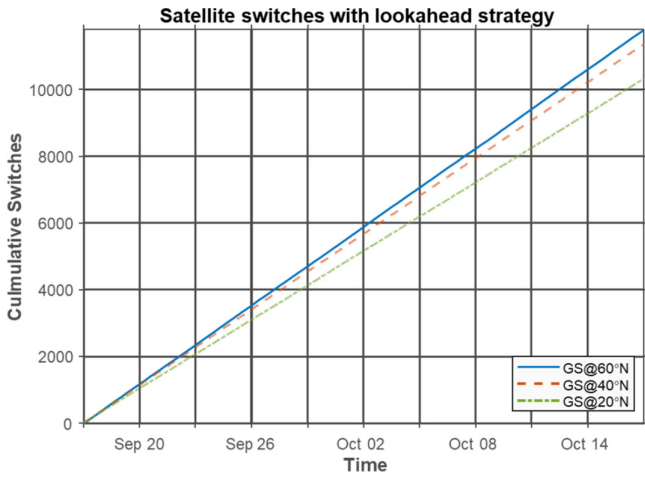


FIGURE 11 | Cumulative switches with lookahead strategy.

Figure 13 summarizes the results of performance in Tables 2, 3, and 4, where HE, LAT, LA denotes highest_elevation, longest_access_time, and lookahead strategies, respectively. These results are precisely aligned with the algorithm's design philosophy. The moderate increase in switches compared to the longest_access_time baseline is the necessary cost of proactively avoiding predicted interference. The lookahead algorithm, guided by the switch penalty, judiciously executes handovers only when the long-term benefit of interference avoidance outweighs the cost of the switch, successfully finding a balance that baseline methods cannot. It prioritizes link quality by eliminating severe interference, while the longest_access_time strategy prioritizes link stability at the cost of accepting periodic interference.

4.2 | Parameter Analysis

To explore the sensitivity of the lookahead algorithm and optimize its configuration, we conducted a parametric analysis evaluating the impact of the lookahead horizon (H), the switching penalty (β), and the discount factor (γ). The analysis in this subsection focuses on the GS at 60°N. Two key performance indicators are analyzed including the average daily handover frequency (measuring link stability) and the probability of severe interference ($P(\alpha < 5^\circ)$) (measuring mitigation effectiveness).

Figures 14 and 15 illustrate the performance results when the discount factor $\gamma = 1$ (i.e., future rewards are undiscounted). Figure 14 plots the average daily satellite switches as a function of the lookahead horizon (H) and the switching penalty (β). Figure 15 plots the corresponding probability of severe interference ($P(\alpha < 5^\circ)$) under the same parameter sweep.

Similarly, Figures 16 and 17 show the results for the same parameter sweep but with the discount factor γ set to 0.5.

The influence of the switching penalty (β) is clear and predictable. Based on the simulation results, for any given lookahead horizon (H), increasing the value of β results in a monotonic decrease in the average number of daily handovers.

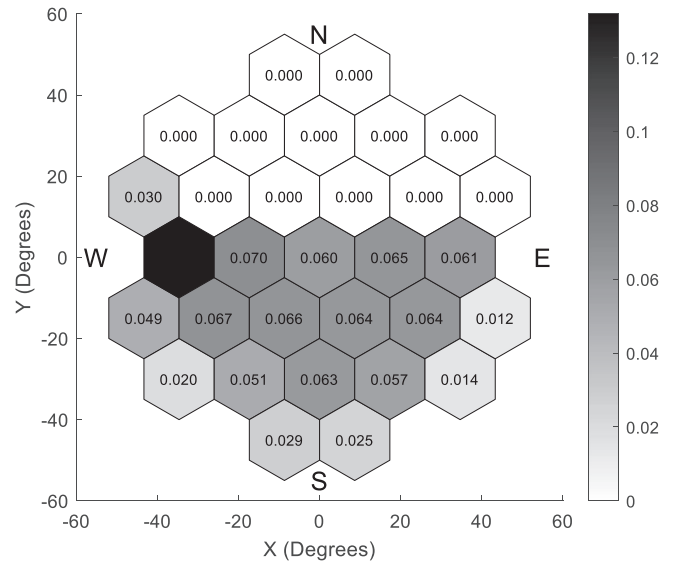


FIGURE 12 | Selection probability with lookahead strategy with GS at 60°N.

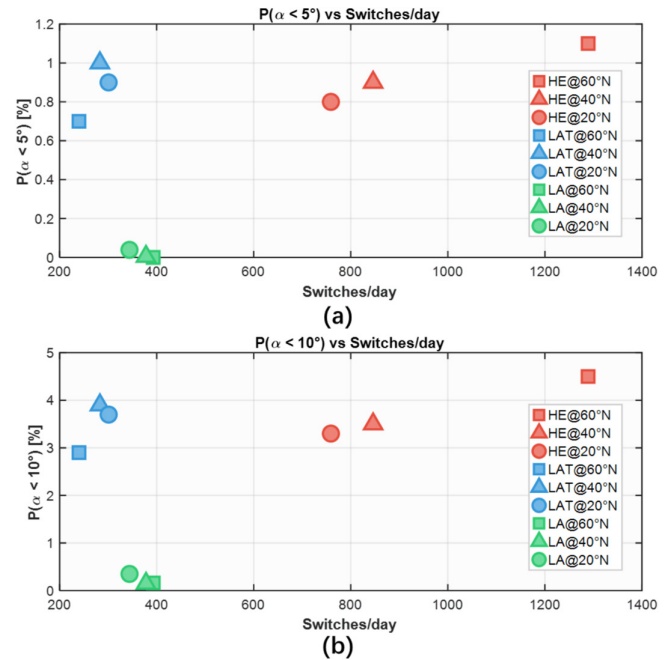


FIGURE 13 | Summary of performance with baseline strategies and lookahead strategies.

This confirms the parameter's role as a direct cost applied to switching, forcing the algorithm to require a significantly better long-term interference profile to justify the instability of a handover.

The lookahead horizon (H) exhibits a more complex, non-linear influence on performance. When analyzing the handover frequency (with $\beta = 0$ to isolate the effect of H), the average daily switches first decrease as H increases from 2, reaching a minimum at $H = 4$, and then begin to increase again for $H > 4$.

Interestingly, the optimal horizon for interference mitigation (the minimum probability of $P(\alpha < 5^\circ)$) is found at $H = 5$ across

various β values. We hypothesize this is linked to the characteristic access duration of the scenario. The longest_access_time baseline, which the victim uses, results in an average connection time of approximately 6.0 min. A 5-min lookahead ($H=5$ with a 1-min step), aligns closely with, but is slightly shorter than, this average access period. This window appears to be optimal for “capturing” the most relevant trajectory information of both the victim and interferer satellites to make a robust decision. In this context, horizon $H < 4$ may be too “myopic,” failing to foresee critical interference events occurring just beyond its planning window. Conversely, a horizon $H > 5$ may be too “long-sighted.” The extended prediction might track satellites into less relevant parts of their pass (e.g., approaching the minimum elevation threshold), diluting the value of the prediction and reducing the effectiveness of the lookahead information.

Finally, the temporal discount factor (γ) acts as a moderator for the switching penalty. As γ decreases, the algorithm places less value on future rewards (at steps $h=1, 2, \dots, H-1$). This makes the immediate penalty C_{switch} more significant relative

to the total discounted future reward. Consequently, for the same H and β , a smaller γ (e.g., 0.5) amplifies the effect of the switch penalty, leading to a greater reduction in handovers as the algorithm prioritizes current-link stability over uncertain future gains.

4.3 | Robustness Analysis: Impact of Orbital Inclination

To assess the robustness of the proposed lookahead algorithm under varying orbital geometries—particularly in scenarios where satellites from the victim and interferer constellations exhibit aligned motion directions, leading to “chasing” effects and extended inline interference durations—we conducted supplementary simulations. These extensions focus on cases where both systems share identical inclinations, thereby maximizing the potential for prolonged co-aligned events.

Two additional configurations were simulated for the study: The first one is the Polar-like Scenario, where both the victim system

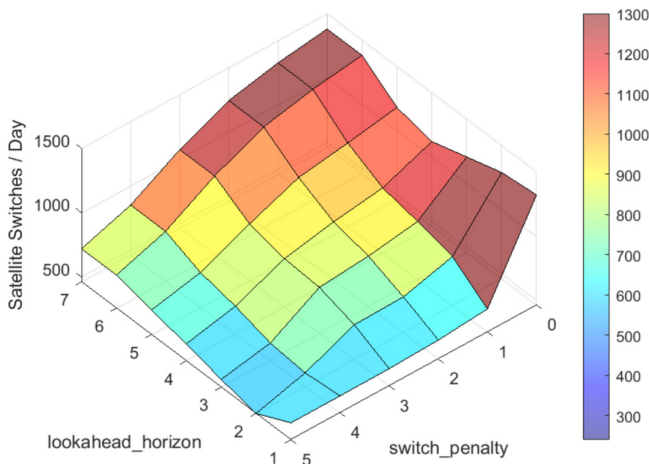


FIGURE 14 | Impact of lookahead horizon (H) and switching penalty (β) on average daily switches (when $\gamma = 1$).

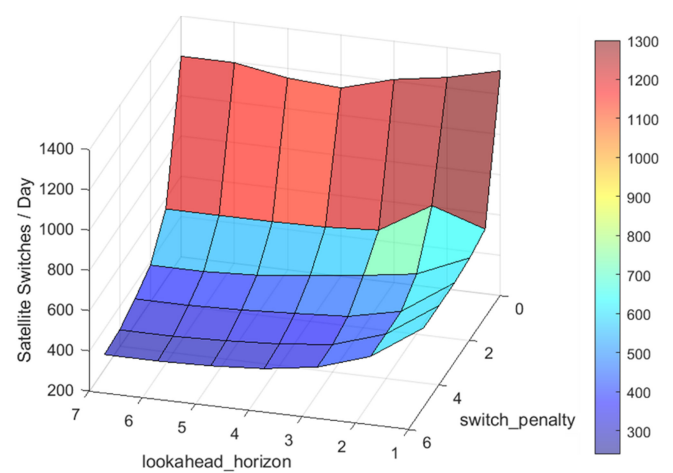


FIGURE 16 | Impact of lookahead horizon (H) and switching penalty (β) on average daily switches (when $\gamma = 0.5$).

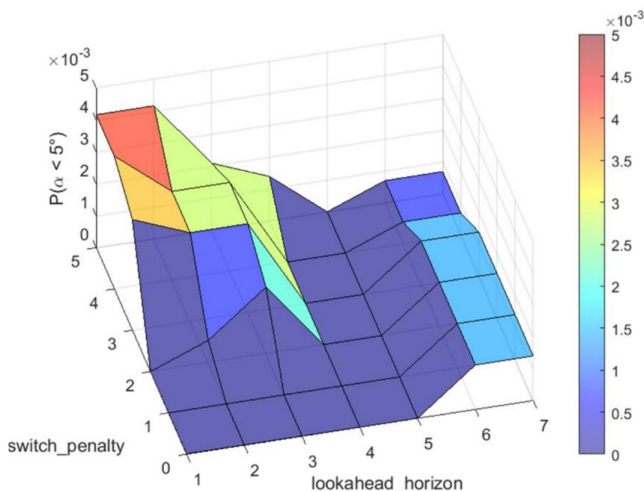


FIGURE 15 | Impact of lookahead horizon (H) and switching penalty (β) on the probability of severe interference ($P(\alpha < 5^\circ)$) (when $\gamma = 1$).

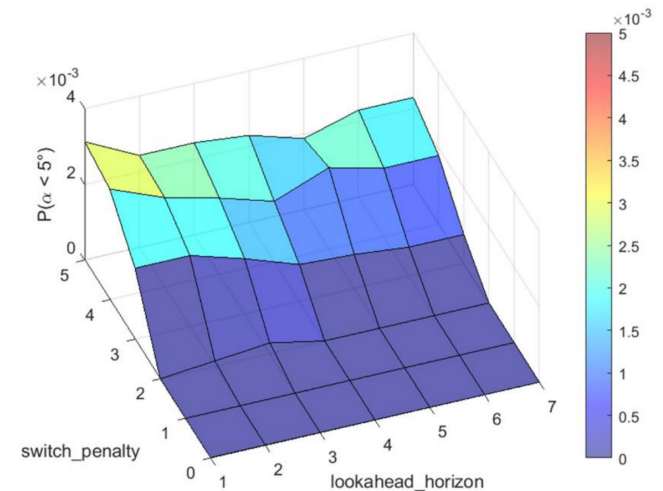


FIGURE 17 | Impact of lookahead horizon (H) and switching penalty (β) on the probability of severe interference ($P(\alpha < 5^\circ)$) (when $\gamma = 0.5$).

and the interferer system are set at an inclination of 88° ; the second is the Inclined Scenario, in which both systems are configured with an inclination of 60° . In both cases, the lookahead horizon H was set to 3, and the switching penalty β was set to 3. This represents a more moderate configuration compared to the primary mixed-inclination evaluations, achieving effective interference avoidance through fewer lookahead steps while maintaining computational efficiency. The results, including cumulative distribution functions (CDFs) of angular separations, are illustrated in Figures 18 and 19, with performance summaries provided in Tables 5 and 6.

The simulation outcomes strongly corroborate the insight that co-inclined orbits exacerbate interference challenges for conventional strategies, while underscoring the resilience of the proposed lookahead approach. For the baseline methods (highest_elevation and longest_access_time), the probability of severe interference ($P(\alpha < 5^\circ)$) markedly increases relative to the original mixed-inclination scenario. Notably, in the $60^\circ/60^\circ$ configuration, this probability escalates to 3.1% for the longest_access_time strategy (compared to $\sim 1.1\%$ in the mixed case), affirming that “chasing” dynamics trap conventional selections in protracted unfavorable geometries.

In contrast, the lookahead strategy exhibits exceptional robustness, achieving severe interference probabilities of 0.012% in the $88^\circ/88^\circ$ scenario and 0% in the $60^\circ/60^\circ$ scenario. This superior performance arises from the algorithm’s multi-step predictive capability, which anticipates and preemptively circumvents these extended “chasing” events by selecting alternative satellites well in advance. Consequently, while identical inclinations amplify the vulnerabilities of baseline approaches, they amplify the value of our proactive method, rendering it indispensable for reliable operation in homogeneous orbital environments.

4.4 | Temporal Granularity Analysis: Impact of Simulation Timestep

To investigate the sensitivity of our approach to temporal resolution—particularly in capturing brief inline events (ILEs) that may span seconds—we extended the simulations to evaluate performance across varying timesteps Δt (60s, 30s, 15s, 6s, 3s). This analysis, conducted at GS latitudes of 60°N , 40°N , and 20°N with $H = 3$ and $\beta = 3$, quantifies how finer sampling affects interference avoidance and handover dynamics. Table 7 summarizes key metrics.

The results indicate that finer Δt slightly reduces $P(\alpha < 5^\circ)$ and $P(\alpha < 10^\circ)$, with marginal, non-monotonic improvements reflecting enhanced detection and preemption of transients. For instance, at GS@ 60°N , $P(\alpha < 5^\circ)$ starts at 0.12% (60s), peaks to 0.19% (30s), then drops to 0.12% (3s); $P(\alpha < 10^\circ)$ rises from 1.01% (60s) to 1.2% (30s) before falling to 0.50% (3s, $\sim 50\%$ net reduction). At GS@ 40°N , $P(\alpha < 5^\circ)$ ranges from 0.0049% (60s) to 0.006% (3s), with minimal change after initial fluctuations; $P(\alpha < 10^\circ)$ decreases from 0.30% (60s) to 0.015% (3s, $\sim 95\%$ improvement). At GS@ 20°N , $P(\alpha < 5^\circ)$ falls from 0.024% (60s) to 0.014% (3s, $\sim 42\%$ reduction); $P(\alpha < 10^\circ)$ from 0.27% (60s) to

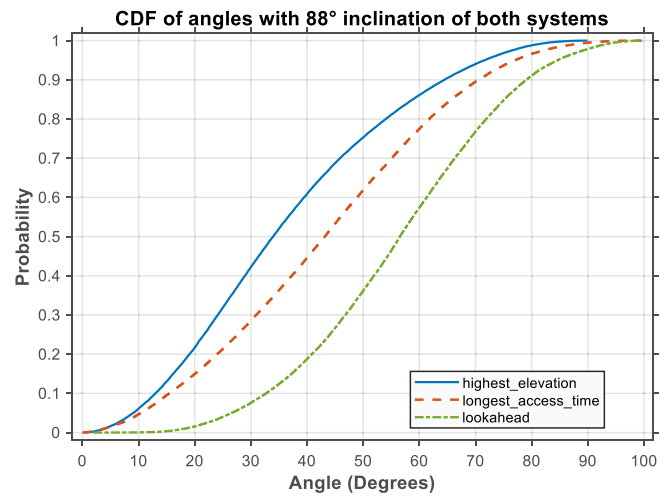


FIGURE 18 | CDF of angles with 88° inclination of both systems.

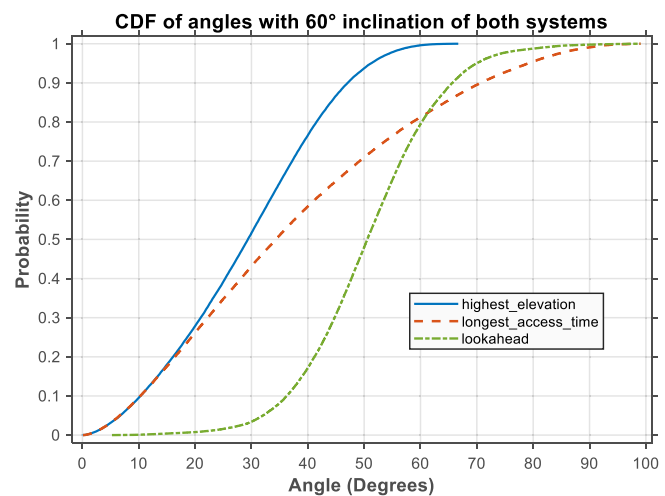


FIGURE 19 | CDF of angles with 60° inclination of both systems.

0.092% (3s, $\sim 66\%$ improvement). Diminishing returns occur below 15s, with P values remaining extremely low, underscoring the lookahead strategy’s robustness.

Min α fluctuates, often dipping at 30–15s before rising (e.g., GS@ 60°N : 1.85° at 60s to 1.25° at 30–15s, then 2.23° at 3s; GS@ 40°N : 4.97° to 1.59° to 4.24° ; GS@ 20°N : 2.66° to 1.32° to 3.93°), indicating finer Δt avoids deeper inline dips. Max α stays stable ($\sim 99^\circ$), with minor variations.

Switches/day increase consistently but plateau at finer Δt (e.g., GS@ 60°N : 368 at 60s to 713 at 3s, $\sim 94\%$ rise, with $< 1\%$ from 6 to 3s; GS@ 40°N : 387 to 585, $\sim 51\%$; GS@ 20°N : 369 to 514, $\sim 39\%$), balancing enhanced reactivity with β -moderated stability. Latitude influences the magnitude, with higher latitudes showing steeper rises due to constrained visibility.

These findings confirm the 1-min baseline’s adequacy for mid-scale constellations while highlighting Δt as a flexible parameter for high-precision applications, with optimal balance around 15–30s where further refinement yields diminishing returns.

TABLE 5 | Performance summary with 88° inclination of both systems.

Strategy	Min α	Max α	P($\alpha < 5^\circ$)	P($\alpha < 10^\circ$)	Switches/day
Highest_elevation	0.086°	89.91°	1.6%	6.1%	585
Longest_access_time	0.086°	99.58°	1.5%	4.7%	334
Lookahead	1.96°	99.58°	0.012%	0.067%	368

TABLE 6 | Performance summary with 60° inclination of both systems.

Strategy	Min α	Max α	P($\alpha < 5^\circ$)	P($\alpha < 10^\circ$)	Switches/day
Highest_elevation	0.078°	66.71°	3.1%	9.5%	1289
Longest_access_time	0.11°	99.08°	3.1%	9.5%	240
Lookahead	5.24°	98.99°	0%	0.088%	317

TABLE 7 | Performance at varying Δt .

	Δt	Min α	Max α	P($\alpha < 5^\circ$)	P($\alpha < 10^\circ$)	Switches/day
GS@60° N	60 s	1.85	99.39	0.12%	1.01%	368
GS@60° N	30 s	1.25	99.52	0.19%	1.2%	518
GS@60° N	15 s	1.25	99.52	0.18%	0.86%	668
GS@60° N	6 s	1.57	99.64	0.15%	0.76%	712
GS@60° N	3 s	2.23	99.65	0.12%	0.50%	713
GS@40° N	60 s	4.97	99.59	0.0049%	0.30%	387
GS@40° N	30 s	1.70	99.36	0.032%	0.29%	498
GS@40° N	15 s	1.59	99.40	0.020%	0.17%	539
GS@40° N	6 s	2.03	99.57	0.008%	0.086%	564
GS@40° N	3 s	4.24	98.68	0.006%	0.015%	585
GS@20° N	60 s	2.66	99.20	0.024%	0.27%	369
GS@20° N	30 s	1.32	99.55	0.021%	0.20%	450
GS@20° N	15 s	1.32	99.47	0.014%	0.14%	491
GS@20° N	6 s	3.93	99.45	0.016%	0.088%	512
GS@20° N	3 s	3.93	99.45	0.014%	0.092%	514

5 | Conclusions

This study addresses the growing challenge of spectrum coexistence in dense NGSO environments by formulating and validating a lookahead-based lookaside satellite selection algorithm. Unlike traditional reactive strategies that optimize instantaneous link metrics, our approach leverages the deterministic nature of satellite orbital dynamics to preemptively avoid future inline interference events. Key conclusions drawn from the simulation and parametric analysis include:

1. **Elimination of severe interference:** The proposed algorithm demonstrates superior interference mitigation capabilities compared to standard baselines. By incorporating multi-step future rewards, it reduced the probability of

severe inline interference (angular separation less than 5°) to near-zero levels across diverse geographic latitudes (0% at 60° N; less than 0.04% at 20° N–40° N), effectively resolving worst-case geometry issues.

2. **Optimized stability-performance trade-off:** The integration of a switching penalty (β) allows the system to balance interference avoidance with link stability. The algorithm successfully curbs the “ping-pong” handover effect observed in highest-elevation strategies, reducing daily switching frequency by approximately 70%, while maintaining sufficient agility to escape deep interference fades that trap longest-access-time strategies.
3. **Robustness in homogeneous orbits:** Supplementary analysis confirms the algorithm's resilience in “chasing”

scenarios where victim and interferer constellations share identical inclinations. While baseline methods failed to mitigate prolonged coaxial tracks in these scenarios, the lookahead logic successfully identified and executed early handovers, maintaining robust isolation.

4. **Parametric design guidelines:** Sensitivity analysis establishes a direct correlation between the optimal lookahead horizon (H) and the characteristic satellite visibility duration. It was observed that a planning window approximating the average pass duration (e.g., 5 min) maximizes interference avoidance efficiency without incurring unnecessary computational overhead.

In summary, this work provides a computationally viable, geometry-based framework for enhancing spectrum sharing in multi-layered satellite networks. Future enhancements to the reward function could incorporate dynamic link budget elements, such as distance-dependent path loss and environmental attenuation factors (e.g., rain fade models from ITU-R recommendations), to more accurately reflect SNR variations over time. This would evolve the current geometry-focused reward into a comprehensive channel state time-series model, enabling finer-grained optimization of both interference avoidance and link quality in diverse operational conditions. Additionally, while the current handover mechanism ensures resource reallocation for the original satellite to non-interfering regions, future work could explore simultaneous multi-satellite service provision to maximize throughput. By leveraging angular separability at the receiver (e.g., through adaptive beamforming or multi-input reception), both victim satellites could potentially serve the GS concurrently, provided isolation angles support signal discrimination. This would require extending the lookahead optimization to joint satellite selection, with rewards incorporating combined SNR and capacity metrics, further enhancing spectrum efficiency in dense NGSO environments.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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